arXiv:0712.4370v1 [physics.ins-det] 28 Dec 2007

The design of the time-of-flight system for MICE

MICE Collaboration

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Abstract. The international Muon Ionization Cooling Experiment (MICE) will carry out a systematic investigation of ionization cooling of a muon beam. As the emittance measurement will be done on a particle-by-particle basis, a sophisticated beam instrumentation is needed to measure particle coordinates and timing vs RF. The MICE time-of-flight system will measure timings with a resolution better than 70 ps per plane, in a harsh environment due to high particle rates, fringe magnetic fields and electron backgrounds from RF dark noise.

The neutrino factory (νF) [1] is a muon storage ring with long straight sections, where decaying muons produce collimated neutrino beams of well defined composition and high intensity. Several νF designs have been proposed, such as the ones of references [2, 3]. The physics program at a neutrino factory is very rich and includes long-baseline ν oscillations, short-baseline ν physics and slow muon physics [4]. The physics performances of a Neutrino Factory depend not only on its clean beam composition ($50\%\nu_e, 50\%\overline{\nu}_\mu$ for the $\mu^+ \mapsto \overline{\nu}_\mu \nu_e e^+$ case), but also on the available beam intensity. The cooling of muons (accounting for $\sim 20\%$ of the final costs of the factory) is thus compulsory, increasing the performances up to a factor 10. Due to the short muon lifetime (2.2 μs), novel methods such as the ionization cooling, proposed more than 20 years ago by A.N. Skrinsky [5], must be used. Essentially the cooling of the transverse phase-space coordinates of a muon beam can be accomplished by passing it through an energy-absorbing material and an accelerating structure, both embedded within a focusing magnetic lattice. Both longitudinal and transverse momentum are lost in the absorber while the RF-cavities restore only the longitudinal component. The MICE experiment [6] at RAL aims at a systematic study of a section of a cooling channel (see figure 1 for a layout). A secondary muon beam from ISIS (140-240 MeV/c central momentum, tunable between 1-12 $\pi\cdot$ mm rad input emittance) enters a cooling section after a diffuser. The 5.5 m long cooling section consists of three absorbers and eight RF cavities encircled by lattice solenoids. The cooling process will be studied by varying the relevant parameters, to allow the extrapolation to different cooling channel designs.

1. The MICE TOF system

Particles are measured before and after the cooling section by two magnetic spectrometers complemented by TOF detectors. For each particle x,y,t, x'=dx/dz=p_x/p_z,y'=dy/dz=p_y/p_z, t'=dt/dz=E/p_z coordinates are measured. In this way, for an ensemble of N particles, the input

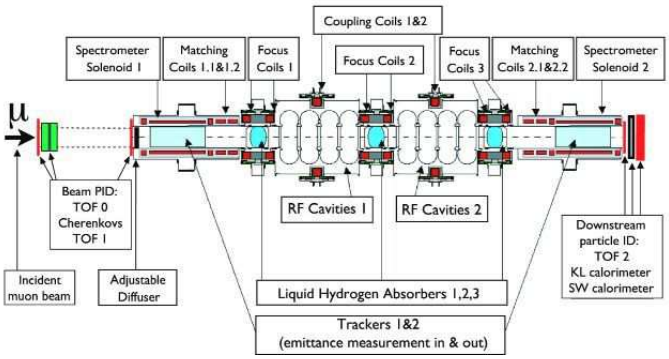


Figure 1. 2-D layout of the MICE experiment at RAL. The beam enters from the left. The cooling section is put between two magnetic spectrometers and two TOF stations (TOF1 and TOF2) to measure particle parameters.

and output emittances are measured with high precision (0.1%). Conventional multiparticle methods cannot be used, due to the correlation between the six particle coordinates, induced by the presence of solenoidal magnetic fields. The driving design criteria of the MICE detector system [7] are robustness, in particular of the tracking detectors, to sustain the severe background conditions in the vicinity of RFs and redundancy in PID in order to keep contaminations (e, π) below 1%. Precision timing measurements are required to relate the time of a muon to the phase of the RF and simultaneously for particle identification by time-of-flight (TOF). A time resolution around 70 ps (σ_t) provides both effective (99%) rejection of beam pions and adequate (5^0) precision of the RF phase. Particle identification is obtained upstream the first solenoid by two TOF stations (TOF0/TOF1) and a Cerenkov counter (CKV1). Downstream the PID is obtained via a further TOF station (TOF2) and a calorimeter, to separate muons from decay electrons. The TOF stations share a common design based on fast 1" scintillator counters along X/Y directions (to increase measurement redundancy) read at both edges by R4998 Hamamatsu photomultipliers [1]. While TOF0 planes cover a 40×40 cm^2 active area, TOF1 and TOF2 cover respectively a 42×42 cm^2 and 60×60 cm^2 active area. The counter width is 4 cm in TOF0 and 6 cm in the following ones. All downstream detectors and the TOF1 station must be shielded against stray magnetic fields (up to 1000-1500 Gauss with a ≤ 400 Gauss longitudinal component, depending on the design of the shielding plates after the spectrometer solenoids). Two options for the local TOF 1/2 shielding are under study: in one a double-sided shielding cage will contain fully the detector, aside an hole for beam, while in the other individual massive soft iron box PMTs shielding are under study [8]. While the first solution is more elegant and reduce the detector weight, it gives complications for detector access and maintenance. Figure 2 shows some preliminary results for the shielding of the most dangerous component of the B field, along the PMT axis, obtained with simple mu-metal and mu-metal+a massive iron box shielding.

Counter prototypes have been tested at the LNF $DA\Phi NE$ Beam Test Facility (BTF) with incident electrons of $E_{beam} = 200 - 350$ MeV to study the intrinsic counter time resolution. The frontend readout used the baseline MICE choice for the TDC: a multihit/multievent CAEN V1290 TDC, in addition to a CAEN V792 QADC (to be replaced in the experiment by CAEN V1724 FADC) for time-walk corrections. The PMT signal was splitted by a passive splitter

[1] 1" linear focussed PMTs, typical gain $G \sim 5.7 \times 10^6$ at B=0 Gauss, risetime 0.7 ns, TTS $\sim$ 160ps

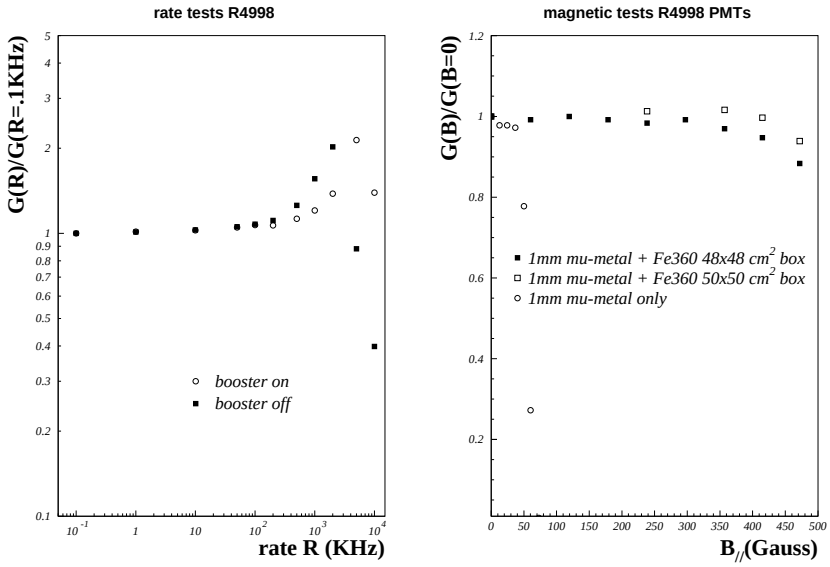


Figure 2. Left panel: rate capability of a typical R4998 PMT, as a function of rate R at field B=0 G (measured P.H. in mV versus rate in KHz). Right panel: tests of shieldings for conventional R4998 Hamamatsu PMTs. The B field is along the PMT axis.

followed by a leading-edge discriminator before the TDC line. An intrinsic single counter resolution $\sim 45 - 60$ ps was obtained depending from beam conditions and the design of lightguides or the used scintillator (Bicron BC404 or BC420 [2], Amcrys-H UPS95F). In the same runs, assuming a gaussian fit for the pulse-height distribution it was possible to estimate the number of photoelectrons per single impinging electron (N_{pe}). From $(< R > /\sigma_R)^2$, where $< R >$ is the peak of the gaussian and σ_R its width, an estimate in the range 200-300 p.e. for BC420 was obtained, depending on the impact point. Clearly, this estimation neglegts electronic noise and is affected by the bad (good) scintillator-PMT coupling.
The TOF stations must sustain a high incoming particle rate (up to 1.5 MHz for TOF0). PMTs rate capabilities were tested in laboratory with a dedicated setup [9] based on a fast laser. An home-made system based on a Nichia NDHV310APC violet laser diode and an AvetchPulse fast pulser (model AVO-9A-C laser diode driver, with ~ 200 ps risetime and a AVX-S1 output module) was used. This system gave laser pulses at ~ 409 nm, with a FWHM between ~ 120 ps and ~ 3 ns (as measured with a 6GHZ 6604B Tek scope) and a max repetition rate of 1 MHz. A typical R4998 PMT had a good rate capability for signals comparable to an incident μ ($\sim$ $300 p.e.$) up to ~ 1 MHz. The rate capability was increased by the use of active bases or a booster on the last dynodes for the R4998 PMTs, as shown in figure 2.

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[2] risetime 0.7 (0.5) ns, $\lambda^{max}_{emission} = 408(391)$ nm, $\lambda^{bulk}_{att} = 160(110)$ cm for BC404 (BC420)